

**Fair representation
is integral to voting
in a Democracy.**

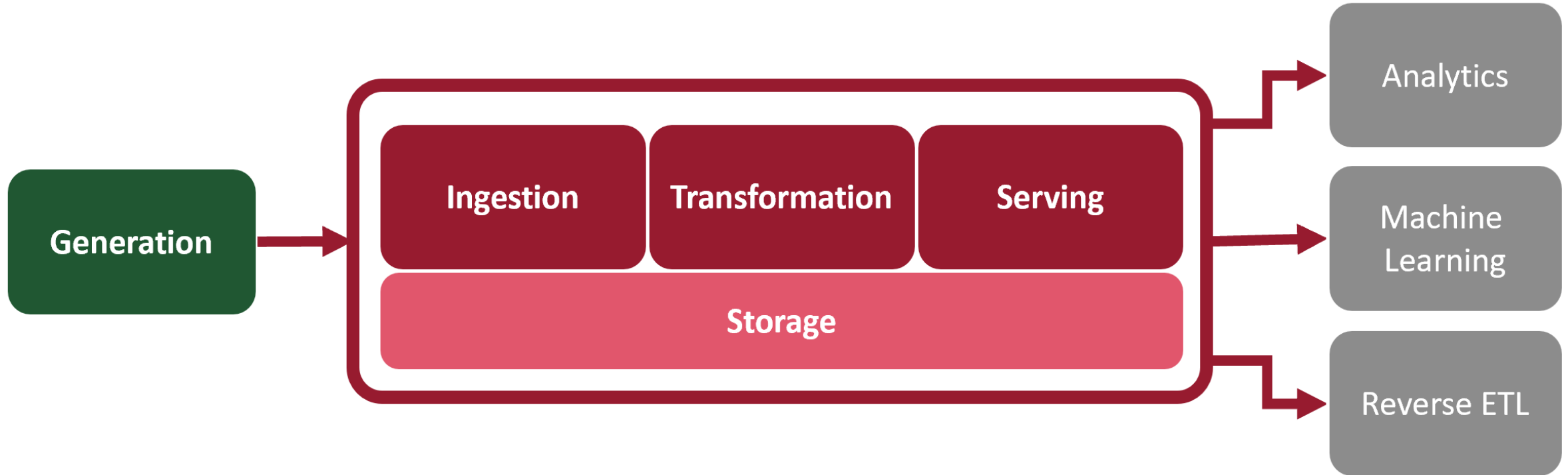
**FAIR
REPRESENTATION**

**Gerrymandering
distorts the
effect of an
individual's vote.**

GERRYMANDERING

Identifying Partisan Gerrymandering Risk in U.S. Congressional Districts

Following The DE Lifecycle



Undercurrents

Security

Data
Management

Data Ops

Data
Architecture

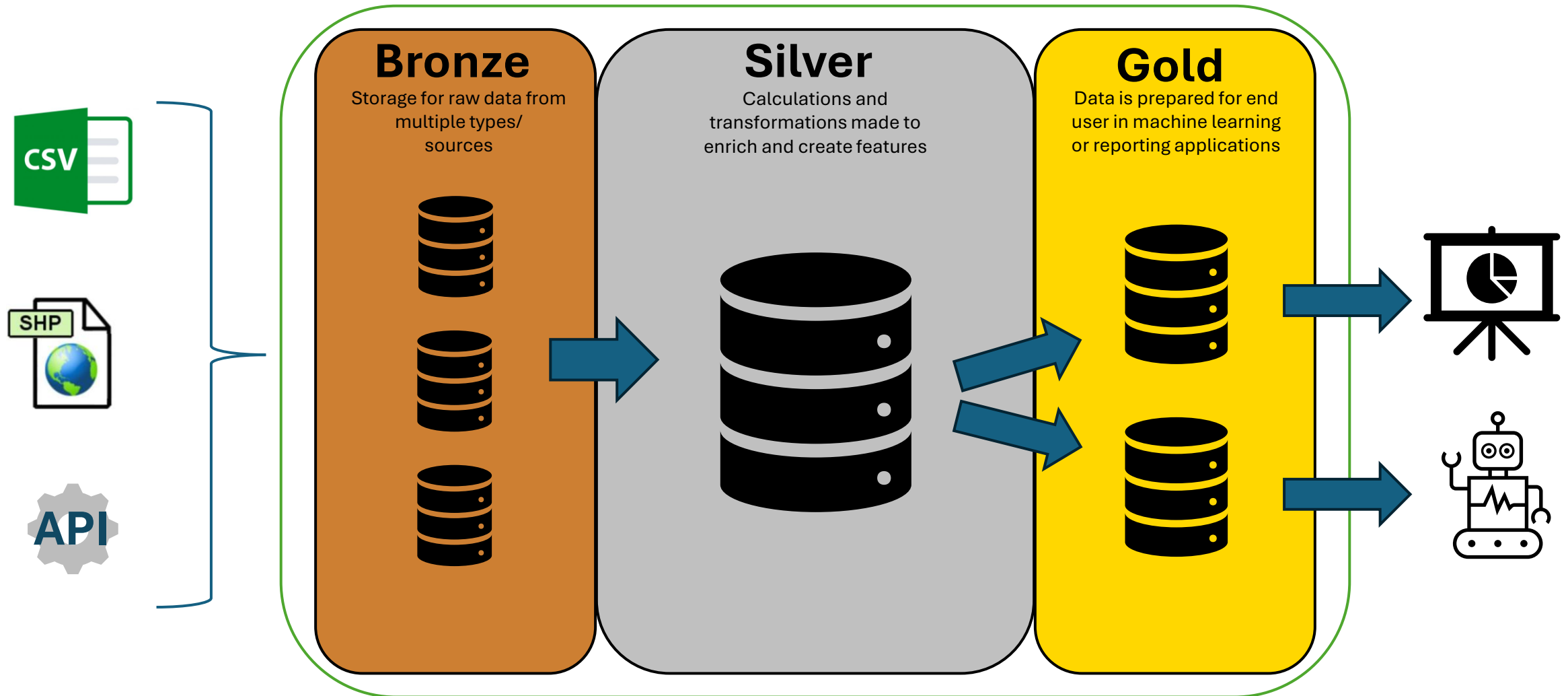
Orchestration

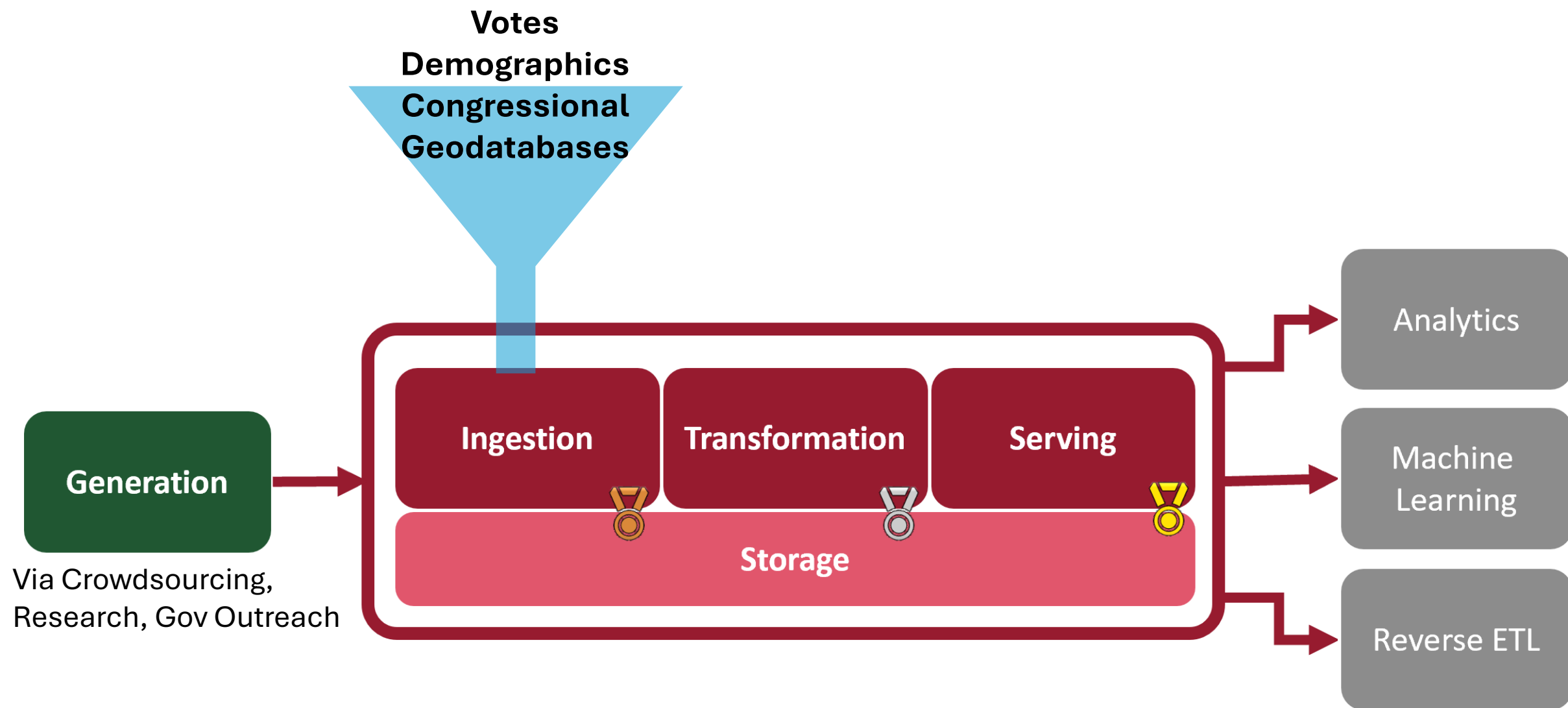
Software
Engineering

Data Sources

Data	Source	Summary	Documentation Process	Accessibility
Voter Data	Open Elections https://openelections.net/	Precinct-level voter data	Crowdsourced	GitHub (CSVs)
Voter Data	Univ. of Florida Election Lab https://election.lab.ufl.edu/precinct-data/	Voter turnout, counts, early voting	Faculty/student generated	Website (Shapefile)
Demographics	Census Bureau – American Community Survey https://www.census.gov/data/developers/data-sets/acs-5year.html	Population, Race, Sex, Income, etc.	Federal government (via US Census)	API
Geographic	Census Bureau – REST Map Services https://tigerweb.geo.census.gov/tigerwebmain/TIGERweb_restmapservice.html	Land use, traits	Federal government	API

Medallion Architecture





Transformation



Filtering and standardizing Open Elections for Congressional data

Calculating voting power, partisan symmetry

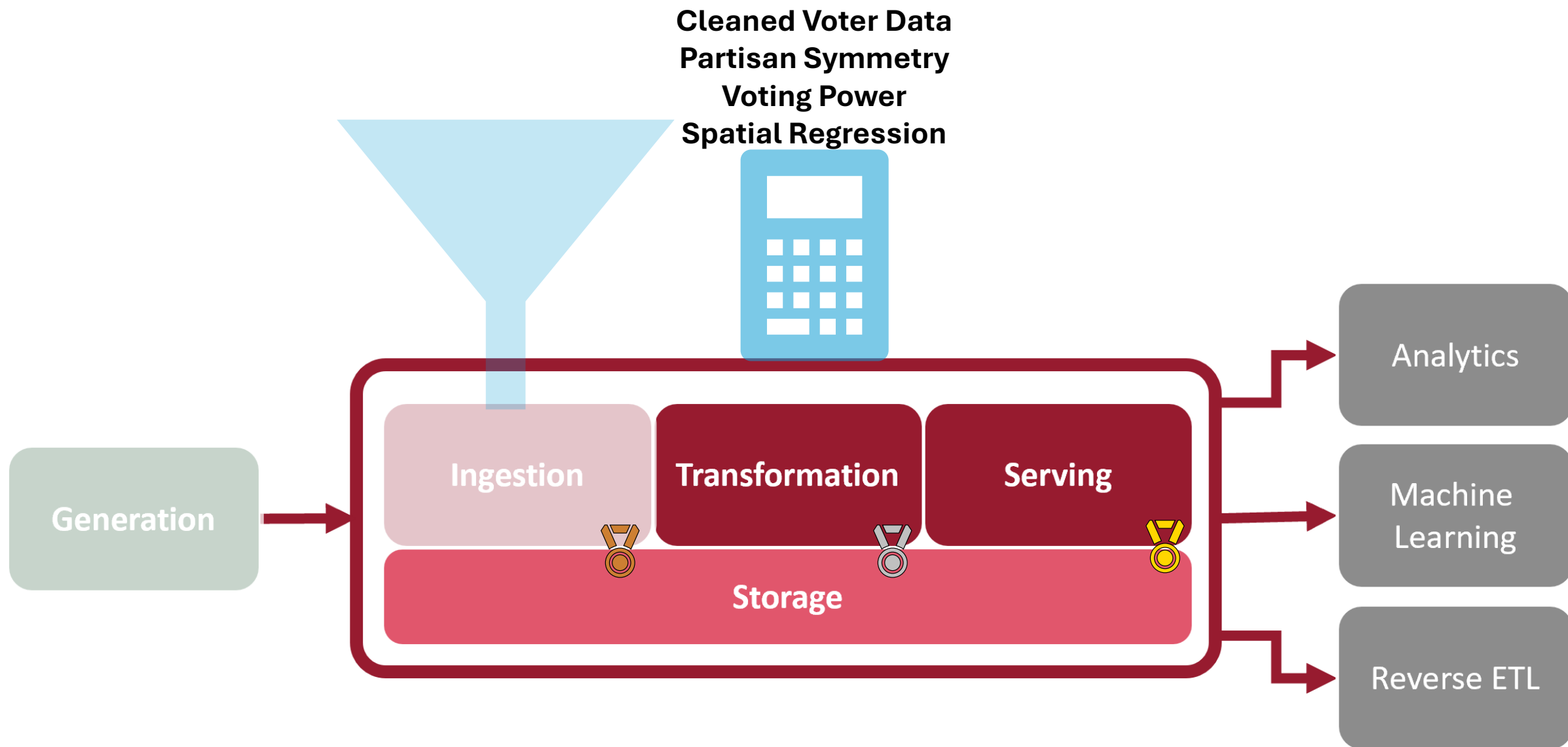
- Seats-Votes Curve, Efficiency Gap, Mean-Median Score

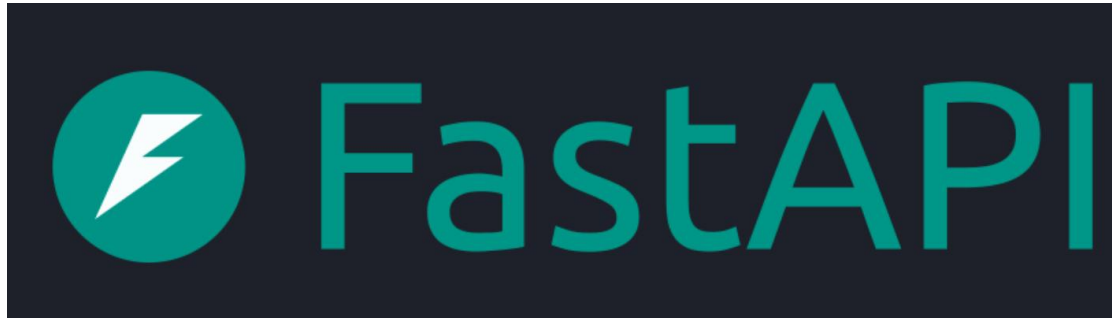
Demographics

- Distribution of populations, populations from precinct to Congressional District

Geospatial calculations

- Spatial Regression – geography as a feature
- Reock Score to measure compactness
- Boundaries (County, City, Precinct) present within a district, crossing districts
- Prior line control (partisan)





Orchestration, Storage, Service

- Cloud storage for source data
- Transform within cloud
- Serve transformed and raw data via API to users for analysis

